Experiment No. 8

Aim: Create a multi-screen app using Fragments and Intents.

Apparatus / Software Required :

Hardware	- Laptop(ASUS Vivobook 16x) - Minimum 8 GB RAM - AMD Ryzen 5 5600H with Radeon Graphics Processor - Internet Connection
Software	- Windows 11 Operating System - Android Studio (Quail) - Java Development Kit (JDK) - Android SDK - Android Emulator (Pixel 7) - Gradle Build System

Theory

A Fragment is a reusable portion of the user interface with its own layout and lifecycle that lives inside an Activity. Using fragments, one Activity can show several screens; the FragmentManager performs transactions (add, replace, remove) to switch fragments inside a container view. Fragments are lighter than activities and are the basis of modern single-activity navigation.

An Intent is a messaging object used to request an action from another component. An explicit Intent names the target class (e.g. DetailActivity.class) and is used inside the app; an implicit Intent declares an action (ACTION_VIEW, ACTION_SEND, ACTION_DIAL, ACTION_SENDTO) and lets the system choose a suitable app. Data is passed using Intent extras (putExtra / getStringExtra) and a result is returned with setResult() and received through the Activity Result API.

Fragments communicate with each other through the host Activity; this project uses a shared ViewModel with LiveData so that the Home fragment can show data submitted from the Form fragment.

Implementation Steps

1. Project Creation: Create an Empty Views Activity named Fragments and Intents using Java and Minimum SDK API 24.

2. Dependencies: Add Material Components and LiveData dependencies in app/build.gradle and sync.

3. UI Design: Create Home, Form, and About fragments, a bottom navigation menu, and a Detail Activity.

4. Fragment Navigation: Use MainActivity to switch between fragments through bottom navigation.

5. Explicit Intent: Validate form inputs and pass name, email, and course details to DetailActivity.

6. Implicit Intents: Implement email sending, website opening, text sharing, and dialer launching.

7. ViewModel and Result: Use a shared ViewModel and LiveData to store and display the submitted result on the Home fragment.

8. Testing: Run the app, navigate between tabs, submit the form, confirm the details, and test all Intent actions.

Files Used / Code Changes

File	Purpose / Change
MainActivity.java	Host activity; bottom navigation and fragment transactions.
HomeFragment.java, FormFragment.java, AboutFragment.java	The three fragments.
DetailActivity.java	Second activity: receives extras, returns result.
SharedViewModel.java	Data sharing between fragments.
activity_main/detail.xml, fragment_*.xml	Screen layouts.
menu/bottom_nav_menu.xml, drawable/ic_*.xml	Navigation menu and its vector icons.
AndroidManifest.xml	Declares MainActivity and DetailActivity.
strings.xml, colors.xml, themes.xml	Resources.
app/build.gradle and other Gradle files	Build configuration.

Components Used

Component	Use in this project
Activity (MainActivity, DetailActivity)	Screens of the app.
Fragment (Home, Form, About)	Sub-screens inside MainActivity.
FragmentManager / FragmentTransaction	Replaces fragments.
BottomNavigationView	Tab navigation.
Explicit and implicit Intents	Navigation, data transfer and calling other apps.
ViewModel + LiveData	Fragment-to-fragment communication.
EditText, Button, TextView, ScrollView	UI widgets.

Important Java Methods Used

Method / Class	Purpose
getSupportFragmentManager().beginTransaction().replace( ).commit()	Swap fragments.
setOnItemSelectedListener()	Handle bottom navigation selection.
onCreateView(), onViewCreated()	Fragment UI creation.
new Intent(context, Class) / putExtra() / getStringExtra()	Explicit intent and extras.
registerForActivityResult(StartActivityForResult)	Receive result from DetailActivity.
setResult(RESULT_OK, data); finish()	Return result.
Intent ACTION_VIEW / SEND / DIAL / SENDTO	Implicit intents.

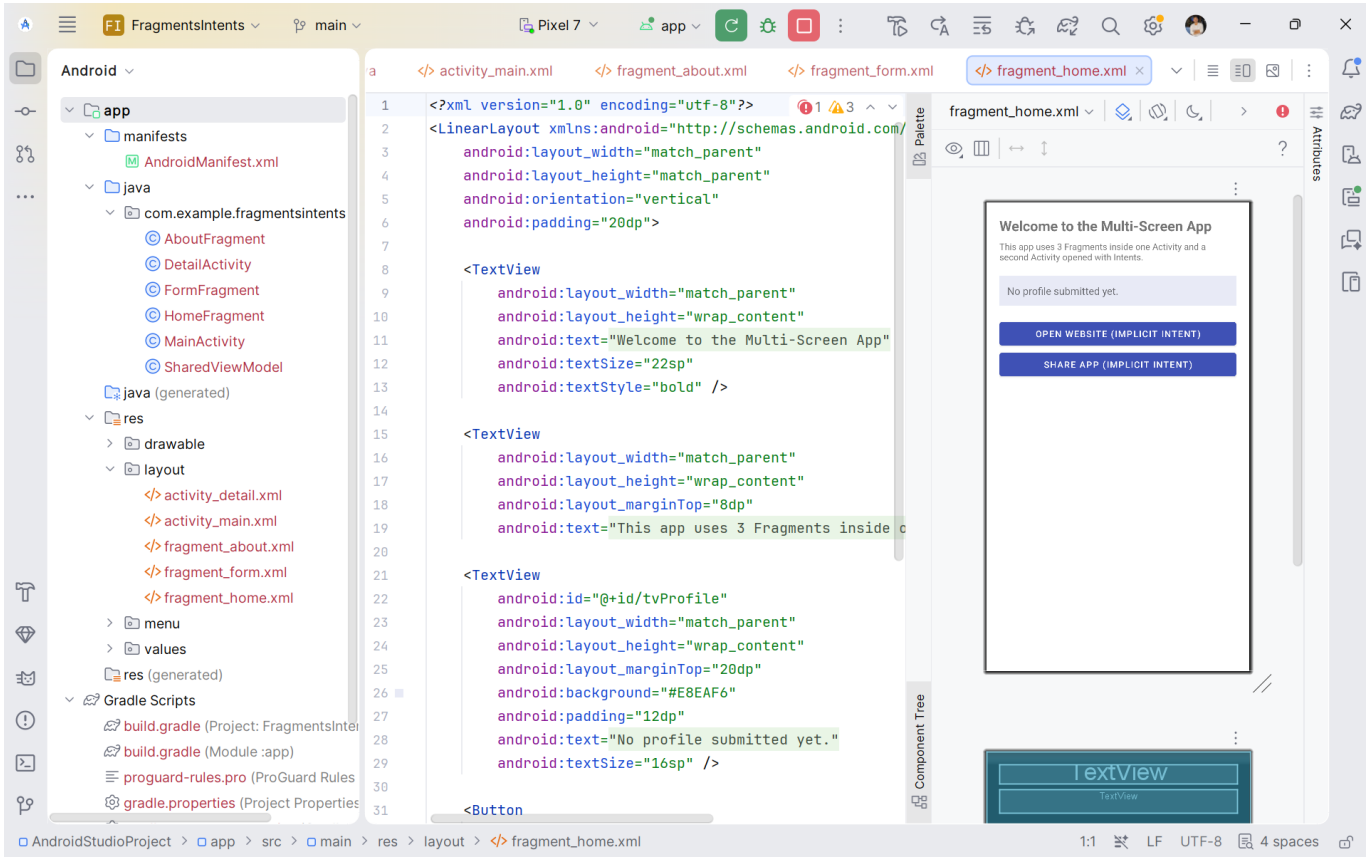
ViewModelProvider(...).get(); LiveData.observe()	Shared state between fragments.
Patterns.EMAIL_ADDRESS.matcher()	Email validation.
EditText.setError()	Show validation message.

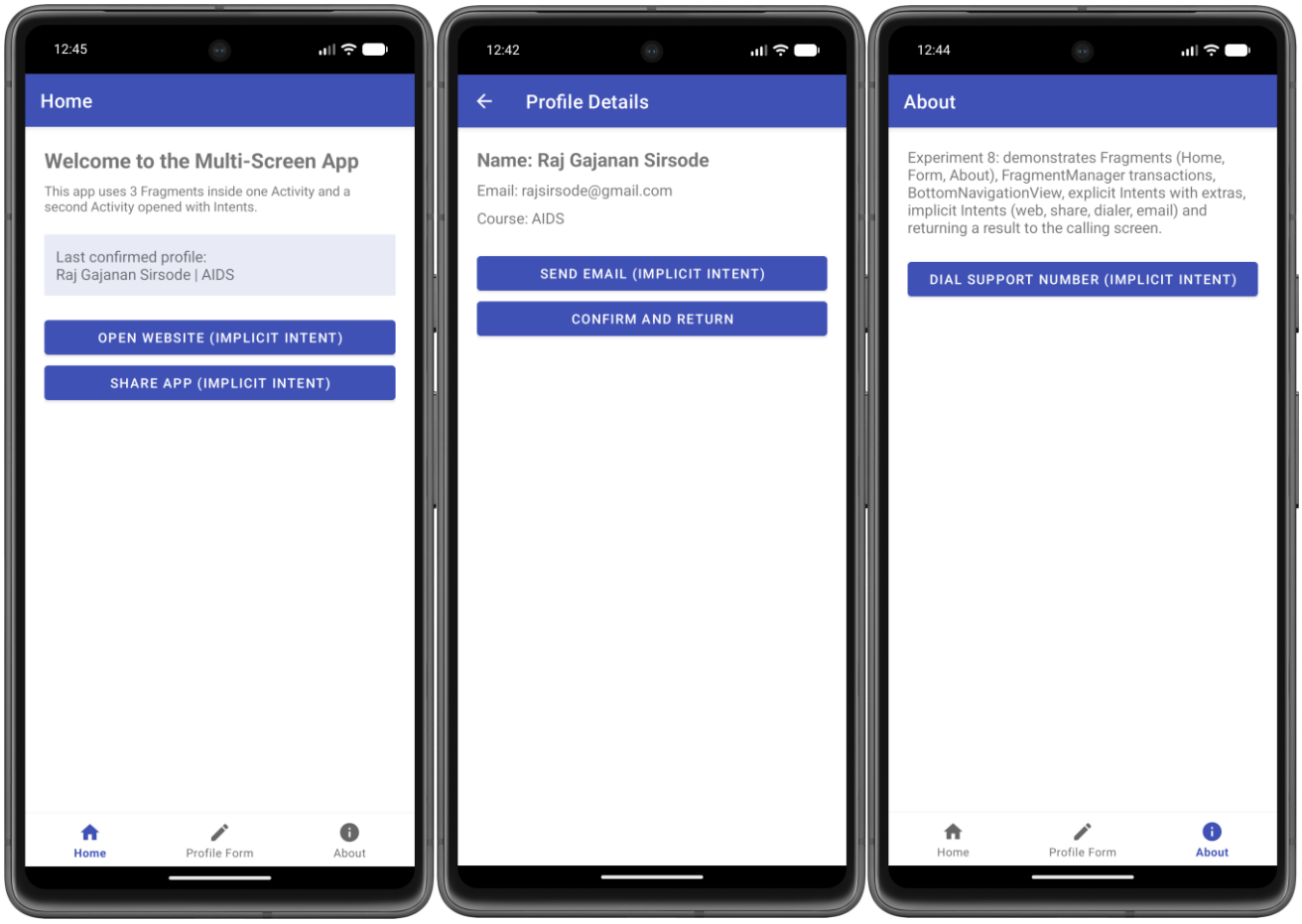
Errors Faced & Troubleshooting

The following are common errors for this experiment and their fixes. Add any additional errors you personally encounter while running it.

Error / Problem	Cause and Fix
ActivityNotFoundException	Activity not declared in manifest, or no app can handle an implicit intent (catch the exception).
Fragment shown twice / overlapping after rotation	Add the first fragment only when savedInstanceState == null.
InflateException for BottomNavigationView	Theme must inherit from Theme.MaterialComponents and Material dependency must be added.
NullPointerException in fragment views	Access views only after onViewCreated().
Result not received	Use the launcher's Intent, call setResult() before finish().
Email / browser does not open on emulator	Emulator has no such app installed; test on a phone.

Output





Conclusion

The experiment demonstrates a multi-screen application built with three Fragments hosted by one Activity and a second Activity reached through explicit Intents, including data passing, returning a result and the use of implicit Intents. The source is complete; build and run it in Android Studio to capture real screenshots.

C (3)	A (3)	P (3)	T (9)	Sign